



Republic of the Philippines

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POULTRYBIZ: AN INTEGRATED FARM MANAGEMENT SYSTEM

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CHAPTER 1: TECHNICAL DESCRIPTION & PROPOSAL

1.1 Background of the Study

The poultry industry is a cornerstone of the Philippine agricultural sector, vital to national food security and rural livelihoods. In the first quarter of 2025, the industry saw a significant 9.3% increase in output, reaching a production value of ₱110.82 billion. This growth is mirrored in the MIMAROPA region and specifically in Marinduque, where chicken inventory surged by 28.3% as of 2023, totaling over one million birds. This upward trend highlights the increasing engagement of local farmers and the immense potential for livelihood development within the province.

Egginear Agri-Poultry Solutions, located in Poras, Boac, Marinduque, is at the forefront of this local growth. Operated by Engr. Don Mark Dela Cruz, the enterprise empowers farmers through science-based egg production systems, providing Ready-to-Lay (RTL) vaccinated pullets bundled with technical training and engineering-based solutions. As a rising agribusiness venture, Egginear currently supports two Sustainable Livelihood Program (SLP) groups and five individual farmers, with plans to onboard more partners across the province.

Despite the industry's modernization in terms of genetics and engineering, a critical bottleneck remains: the reliance on manual recordkeeping systems. Smallholder farmers and SLP groups partnered with Egginear still depend on traditional logbooks and physical paper trails to track egg inventory, sales, expenses, and flock health. These manual processes are inherently:

- **Time-Consuming:** Farmers spend significant time manually consolidating daily records from various barns.
- **Error-Prone:** Physical logs are susceptible to human error, data loss, or physical damage (e.g., moisture or dirt in the farm environment).
- **Difficult to Analyze:** Without a centralized digital system, it is challenging for the owner to monitor real-time productivity or identify early signs of health issues through mortality rates.

This inefficiency hinders data-driven decision-making and limits the overall productivity and sustainability of poultry operations. There is a pressing need for a digitized management system—PoultryBiz—to bridge this gap, ensuring that the engineering-based efficiency championed by Egginear is supported by an equally advanced and reliable digital infrastructure.

1.2 Objectives

General Objective: To design, develop and deploy PoultryBuz, an integrated web farm management system for Egginear Agri-Poultry Solutions that automates recordkeeping and provides real-time data analytics for egg production, inventory, and financial monitoring by the end of the end of the current semester.

Specific Objectives:

1. To develop a high-performance frontend interface using React.js and Vite, providing a fast, responsive, and optimized user experience for both the owner and farm staff.
2. To develop a secure user authentication and authorization module using JSON Web Tokens(JWT) and Role-Based Access Control(RBAC) that separates access privileges among Admin, Farmers, and Visitors.
3. To implement a centralized web dashboard for real-time monitoring of egg production, sales, and expenses, utilizing React Router for seamless, multi-page navigation.
4. To create a robust Backend API using Node.js and Express.js to manage complex business logic and facilitate secure data communication between the client and the database.
5. To build a responsive and user-friendly web-based interface that enables authorized users to access the system using desktop browsers within the farm environment.
6. To develop a cloud-based database solution using MongoDB Atlas and Mongoose for flexible, scalable modeling of poultry inventory, mortality logs, and feed usage.
7. To ensure system reliability and data integrity by performing comprehensive endpoint testing via Postman and Thunder Client to verify API security and performance.

1.3 Review of Related Literature and Systems

Related System 1: Smart Poultry Farming (Mobile-based IoT System)

This system solved the problem of manual monitoring by implementing environmental sensors and a mobile application for real-time detection of feed and water levels. However, it is heavily dependent on stable internet connectivity and expensive hardware sensors, which is a significant barrier for smallholder farmers in Marinduque. PoultryBiz addresses this by focusing on a robust, mobile-responsive recordkeeping system that can operate effectively even with limited hardware integration and can be adapted for offline data entry.

Related System 2: Integrated Farming System (IFS) Models

These models improved livelihood security by recycling farm resources (manure to crops/fish) and increasing net returns for smallholders. While effective in resource management, the system is highly region-specific and lacks a digital platform for daily operational tracking. PoultryBiz addresses this by providing a digital

infrastructure that allows farmers to monitor these integrated activities specifically for egg production, making the management process more systematic and less reliant on manual estimation.

Related System 3: Mechatronic and Automation Technology for Sustainable Farming

This conceptual system proposed the use of AI, drones, and RFID for livestock and poultry monitoring to improve production efficiency. Its main limitation is that it remains primarily descriptive and too costly for actual field implementation in developing rural areas. PoultryBiz addresses this by providing a practical, low-cost software solution that utilizes the farmer's existing smartphones, focusing on immediate operational needs like inventory and sales rather than expensive robotics.

Related System 4: Network-based Digital Transformation Model

This system solved the problem of project prioritization and resource allocation in medium-sized poultry farms using a network approach. Its limitation is its focus on high-level managerial contexts, making it too complex and inapplicable for small-scale individual farmers or SLP groups. PoultryBiz addresses this by simplifying the user interface and focusing on daily farm-level tasks, ensuring that the technology is accessible to users regardless of their enterprise size or managerial background.

Related System 5: Social Media Analytics for Integrated Cropping Systems

This system used digital platforms to identify informational needs and promote the adoption of poultry practices among farmers. However, it functions only as a communication and extension tool and does not provide an actual technical system for managing farm data. PoultryBiz addresses this by moving beyond mere information sharing and providing a dedicated management application where farmers can directly input, store, and analyze their own farm's productivity data.

1.4 Methodology & Solution

This study outlines the technical framework and procedural approach to be utilized in the development of PoultryBiz. It adopts the Agile Development Model, characterized by iterative cycles of planning, executing, and testing, to ensure the system remains responsive to the dynamic operational requirements of poultry farming.

The system is developed using the MERN Stack, consisting of MongoDB, Express.js, React, and Node.js. MongoDB serves as a flexible NoSQL database capable of handling complex and dynamic agricultural data. Express.js functions as the backend framework, managing routing and server-side logic. React is used to build a responsive and dynamic user interface, enabling real-time updates without page reloads. Node.js acts as the runtime environment, supporting efficient handling of multiple concurrent operations.

The system follows a client-server architecture utilizing RESTful APIs. The backend processes data requests, enforces security through JWT authentication, and communicates with the database. Meanwhile, the frontend

consumes these APIs to display real-time data through an interactive dashboard, allowing users to monitor farm operations efficiently.

1.4.1 The MERN Stack Solution

The researchers will utilize the MERN Stack (MongoDB, Express.js, React, and Node.js) as the core technical foundation. This full-stack JavaScript solution was selected for its high performance, scalability, and seamless data flow between layers.

- MongoDB (Database Layer, as a NoSQL, document_oriented database. MongoDB provides the necessary schema flexibility to store complex agricultural data. Unlike traditional relational databases, MongoDB allows for the storage of hierarchical data such as varying flock batches, multifaceted logs, and non-uniform transaction histories. This ensures that the system can evolve without the need for frequent and disruptive database migrations.
- Express.js (Web Application Framework), functioning as the backend framework. Express.js provides a robust set of features for web and mobile applications. It acts as the middleware that manages the server's routing logic, ensuring that client-side requests from the farm's frontend are processed with minimal latency.
- React (Frontend Layer), this component-based library is utilized to build a highly responsive User Interface (UI). Through its Virtual DOM (Document Object Model) implementation, React enables real-time updates of the farm's dashboard such as instantaneous inventory level changes without requiring a full page refresh. This provides farm personnel with a seamless, application-like experience across desktop and mobile devices.
- Node.js (Runtime Environment), serving as the engine for the server-side. It utilizes an event-driven, non-blocking I/O model. This is critical for poultry management systems that may need to handle multiple concurrent data streams, such as simultaneous inputs from inventory updates and high-volume sales transactions, without compromising system stability.

1.4.2 System Architecture and API Consumption

The system adheres to a Client-Server Architecture utilizing RESTful APIs (Representational State Transfer). The architectural flow is divided into two primary phases:

- (1) Back-end Logic and Data Transformation - The Node.js and Express.js server communicates with MongoDB to perform CRUD (Create, Read, Update, Delete), which acts as a lightweight, language-agnostic data format. Security is enforced at this layer using JWT (JSON Web Tokens) to ensure that only authorized farm personnel can access sensitive financial and production records.

(2) Front-end State Management and Consumption - The React frontend consumes the backend APIs using the Axios library. To maintain a consistent view of the farm’s status, the system utilizes React Hooks (e.g., useState and useEffect) for local state management. This ensures that whenever a change occurs in the backend-such as depletion in feed stock, the frontend immediately reflects the most current data, facilitating proactive decision-making.

1.4.3 Quality Assurance and Testing Strategy

To guarantee the reliability and accuracy of the system in a live agricultural setting, a rigorous three-tiered testing strategy will be implemented:

1. Unit Testing - This focuses on the smallest testable parts of the application. The researchers will use Postman/Thunder Client to verify the accuracy of individual logic functions, such as those calculating Feed Conversion Ratios (FCR), daily mortality rates, and automated sales totals.
2. Integration Testing - This phase validates the interaction between different system modules. It ensures that the communication between the React frontend and the Express/Node backend remains stable and that data sent from the UI is correctly persisted in the MongoDB database without corruption.
3. End-to-End (E2E) Testing - Using Cypress, the researchers will simulate complete user journeys, such as “Batch Arrival to Market Sale”. This ensures that the entire system works harmoniously from the user’s perspective, covering authentication, data entry, and report generation, thereby ensuring the system’s readiness for actual deployment at Egginear Agri-Poultry Solutions.

1.5 Target Users/Beneficiaries

This system aims to provide poultry farm stakeholders with organized, accurate, and secure access to their operational data.

Beneficiary 1: Farm Owner/System Administrator (Primary User)

The Farm Owner or System Administrator holds the highest access, managing user accounts, assigning roles, monitoring farm operations, accessing all records and reports, overseeing inventory, egg production, and health records of the chickens, configuring settings, and ensuring data integrity.

Beneficiary 2: Farmers/Staff (Secondary User)

Farmers or Staff are responsible for inputting daily operational records into the system. These records include feed distribution, egg production, mortality incidents, cleaning activities, vaccination completion, and flock

observations. To maintain the integrity of sensitive data, access privileges will be restricted according to each staff member's specific responsibilities, preventing unauthorized modifications.

1.6 Significance of the Study

The development of PoultryBiz: An Integrated Farm Management for Egginear Agri-Poultry Solutions is expected to provide significant benefits to various stakeholders by addressing the operational gaps in traditional poultry farming. The importance of this study is highlighted through the following:

To the Farm Owners: The system provides a centralized platform for real-time monitoring of farm activities, sales, and inventory. It eliminates the risks of data fragmentation and feed leakage inherent in manual record-keeping, leading to better financial transparency and increased profitability through data-driven decision-making.

To the Farmers: By automating daily logs and inventory tracking, the system reduces the administrative burden and human error associated with paper-based documentation. This allows the staff to focus more on technical husbandry tasks and the physical welfare of the poultry, thereby improving overall operational efficiency.

To Future Researchers: The findings and the technical architecture of this project can serve as a foundational reference for future studies involving agricultural automation, digital diaries, and the integration of sales-inventory systems in different livestock sectors.

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